ETH zürich

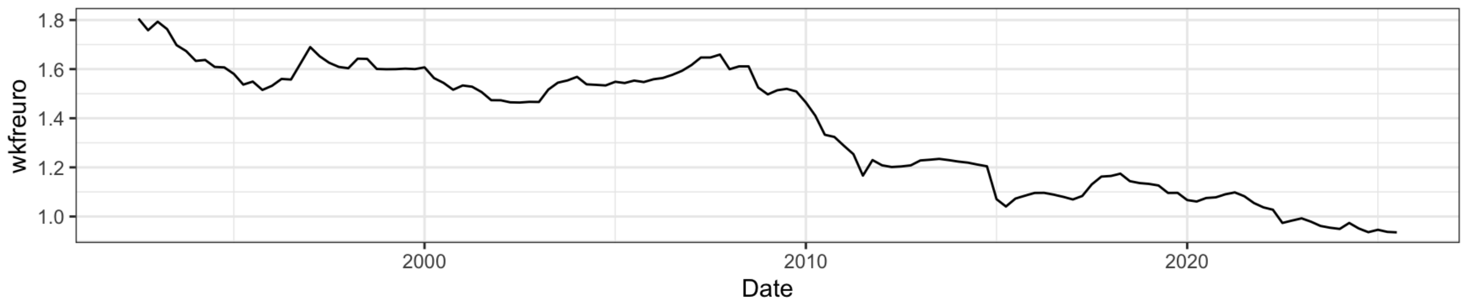
Forecasting with Bayesian VAR

Giacomo Maggiore, Ben Melcher

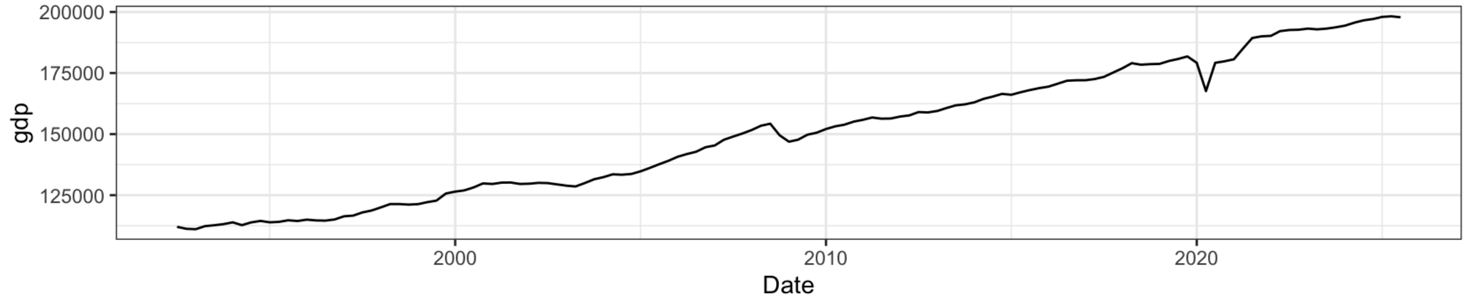
Methods in Macroeconomic Forecasting 2025
19. Novembeer 2025, Zurich

Variables to Forecast

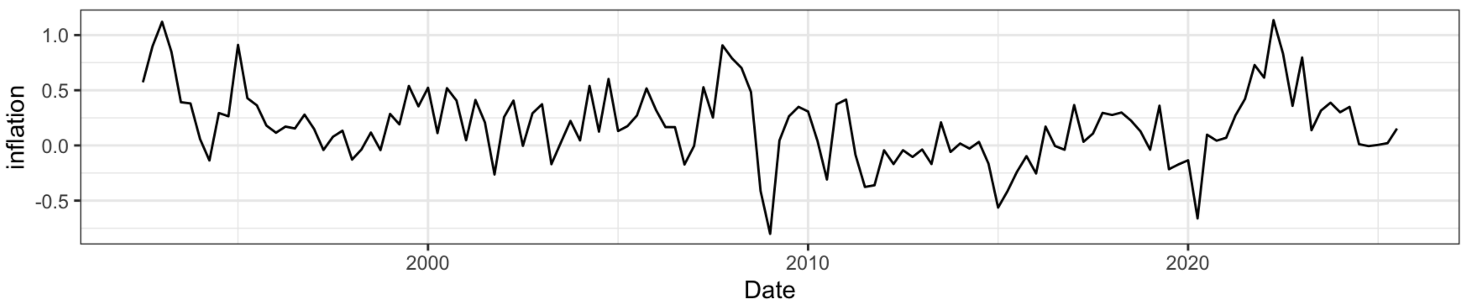
Exchange Rate



GDP



Inflation Rate



Theory Review

VAR (Vector Autoregression)

Features

- Models the joint dynamics of multiple macroeconomic variables.
- All variables are treated as endogenous.
- Each variable depends on its own past values and the past values of all other variables.
- Widely used for short-to-medium-term macroeconomic forecasting and shock transmission analysis.

$$\mathbf{y}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{A}_i \mathbf{y}_{t-i} + \epsilon_t$$

Coefficients are typically estimated using Ordinary Least Squares (OLS) applied separately to each equation in the system.

```r
library(vars)
library(dplyr)

df <- utils::read.csv("data/data_quarterly.csv")
df$date <- as.Date(paste0(df$date, "-01"))

selected_variables <- c("gdp", "wkfreuro", "inflation")

df <- df %>%
  filter(!is.na(gdp), !is.na(wkfreuro), !is.na(inflation)) %>%
  select(date, all_of(selected_variables))

lag_selection <- VARselect(df %>% select(-date), lag.max = 8, type = "const")
optimal_lag <- lag_selection$selection["AIC(n)"]

var_model <- VAR(df %>% select(-date), p = optimal_lag, type = "const")

forecast_horizon <- 4
var_forecast <- predict(var_model, n.ahead = forecast_horizon, ci = 0.95)
```

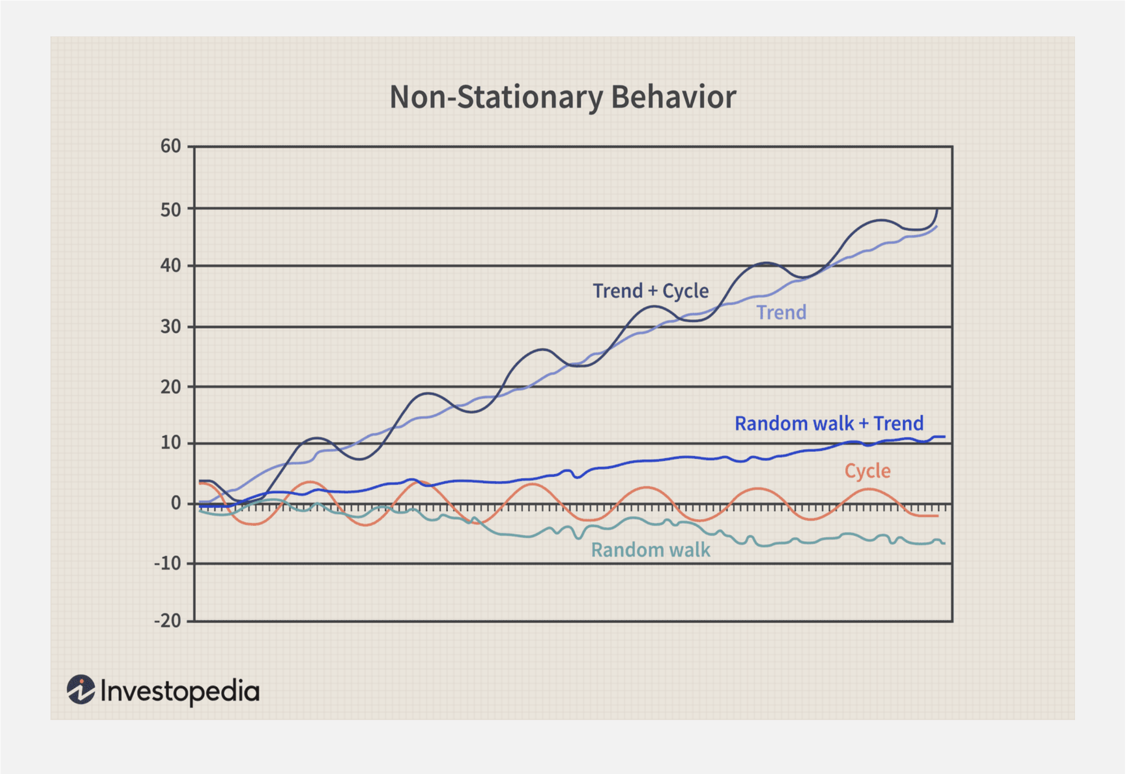
Testing for Stationarity (The ADF Test)

ADF Test is a popular formal statistical test used to check if a unit root* is present in a time series, which is the definition of non-stationarity.

Hypotheses:

- **Null Hypothesis (Ho):** The time series **is non-stationary** (a unit root exists).
- **Alternative Hypothesis (Ha):** The time series **is stationary**.

*If the resulting p-value is less than the significance level (e.g., 0.05), we reject **Ho** and conclude the series is likely stationary.*

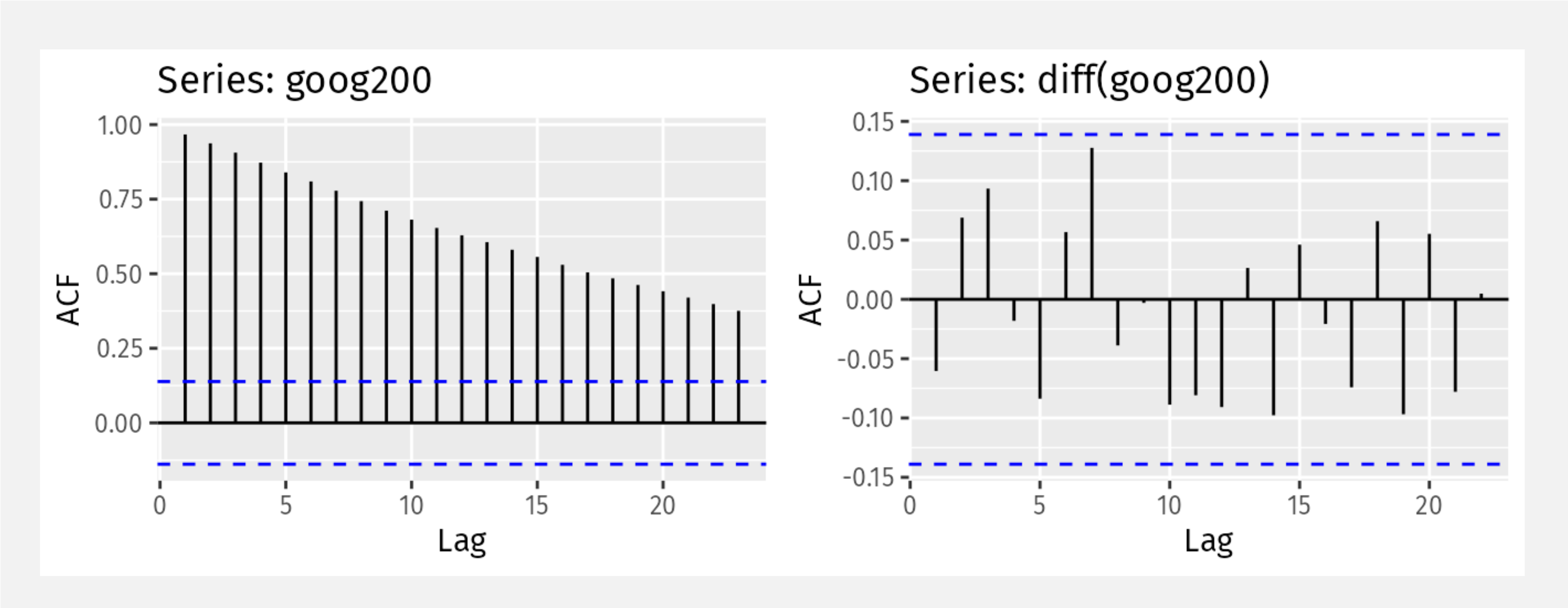


***unit root: stochastic trend in a time series** that is frequently referred to as a random walk with drift.

Data Stationarity

A time series is **stationary** if its statistical properties (mean, variance, and autocorrelation structure) do not change over time.

Many traditional time series models assume stationarity to ensure reliable parameter estimation and forecasting.

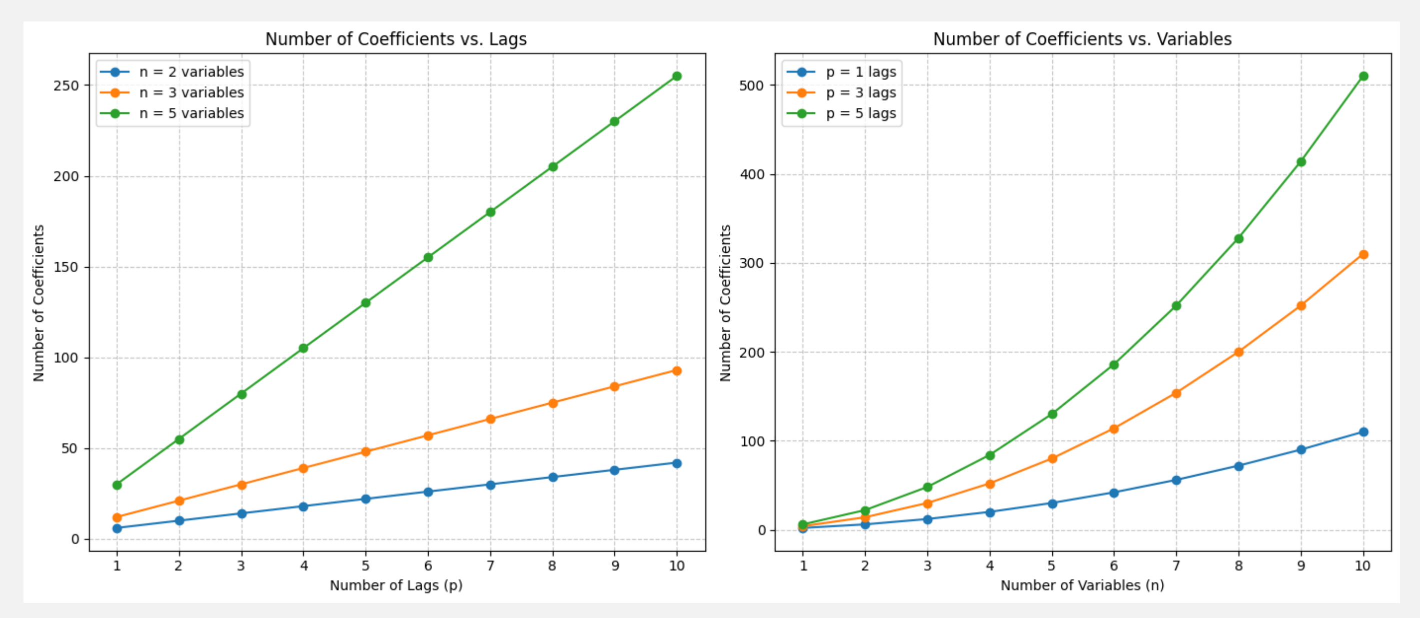


Key Problems and Limitations

- **Overparameterization**: With many variables and lags, the number of parameters explodes leading to risk of overfitting.
- **Poor performance with small samples**: Macroeconomic data are often limited, making OLS-estimated VARs unstable.
- **Multi-collinearity**: Highly correlated predictors lead to noisy coefficient estimates.
- **Weak long-horizon forecasts**: VARs often revert too quickly to unconditional means.

*For a VAR with **n variables** and **p lags**:*

$$\text{Number of coefficients} = n \times (n \times p + 1)$$



Bayesian VARs

A VAR model estimated using **Bayesian methods**, which combine observed data with "prior beliefs" (priors) about the parameters.

Priors are statistical assumptions that "shrink" the model that shrink coefficients toward zero and reduce estimation noise.

BVAR returns a density forecasts by construction: Bayesian methods naturally deliver predictive distributions.

From the Bayes Theorem

$$p(\Theta|Y) \propto \mathcal{L}(Y|\Theta) \cdot p(\Theta)$$

Where:
- $p(\Theta|Y)$ is the Posterior Distribution.
- $\mathcal{L}(Y|\Theta)$ is the Likelihood Function of the data Y given the parameters Θ.
- $p(\Theta)$ is the Prior Distribution of the parameters Θ.

Possible Priors

- **Diffuse / Flat Priors:** no shrinkage → equivalent to OLS
- **Normal–Inverse Wishart :** conjugate prior → fast analytical posterior, moderate shrinkage.
- **Independent Normal × Inverse Wishart:** more flexibility; often used in modern applied BVARs.
- **Minnesota Prior:** Strong shrinkage toward random walks.

Literature shows that informative priors (e.g., Minnesota) reduce estimation errors and improve forecast accuracy compared to flat priors

BVAR: Bayesian Vector Autoregressions with Hierarchical Prior Selection in R

Nikolas Kuschnig
WU Vienna University
of Economics and Business

Lukas Vashold
WU Vienna University
of Economics and Business

Abstract

PRIOR SELECTION FOR VECTOR AUTOREGRESSIONS

Domenico Giannone, Michele Lenza, and Giorgio E. Primiceri*

Abstract—Vector autoregressions (VARs) are flexible time series models that can capture complex dynamic interrelationships among macroeconomic variables. However, their dense parameterization leads to unstable inference and inaccurate out-of-sample forecasts, particularly for models with many variables. A solution to this problem is to use informative priors in order to shrink the richly parameterized unrestricted model toward a parsimonious naıve benchmark, and thus reduce estimation uncertainty. This paper studies the optimal choice of the informativeness of these priors, which we treat as additional parameters, in the spirit of hierarchical modeling. This approach, theoretically grounded and easy to implement, greatly reduces the number and importance of subjective choices in the setting of the prior. Moreover, it performs very well in terms of both out-of-sample forecasting—as well as factor models—and accuracy in the estimation of impulse response functions.

effectively reduce the estimation error while generating only relatively small biases in the estimates of the parameters. For a more formal illustration of this point from a Bayesian perspective, we consider the following (conditional) prior distribution for the VAR coefficients,

$$\beta|\Sigma \sim N\left(b, \Sigma \otimes \Omega\xi\right),$$

where the vector b and the matrix Ω are known and ξ is a scalar parameter controlling the tightness of the prior information. The conditional posterior of β can be obtained by multiplying this prior by the likelihood function. Taking the initial p observations of the sample as given, a stan-

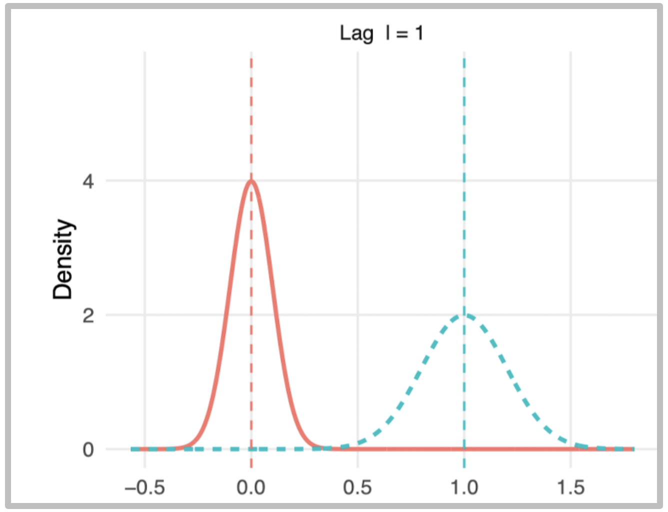
Deep Dive: The Minnesota Prior

Key Assumption: a variable's own past is the best predictor of its future (a random walk).

The prior is built on the belief that the best predictor of a variable is its own most recent value:

- *Prior Mean is 1 (for a variable's own first lag)*
- *Prior Mean is 0 (for all other coefficients)*

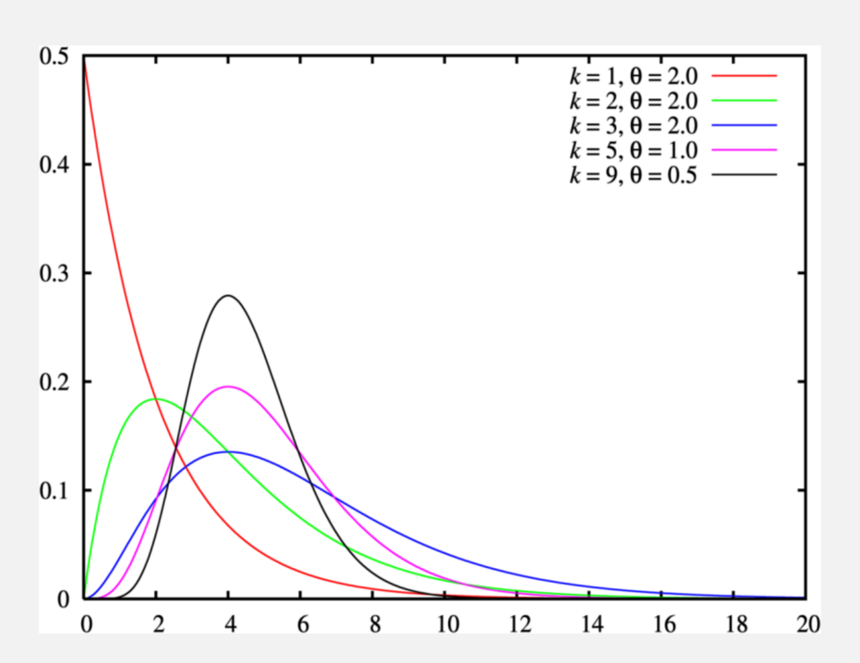
$$\mathbb{E}[(\mathbf{A}_s)_{ij}|\mathbf{\Sigma}] = \begin{cases} 1 & \text{if } i = j \text{ and } s = 1, \\ 0 & \text{otherwise.} \end{cases}$$



*The strength of this belief is decided using a **"tightness" hyperparameter (λ).***

Hierarchical Prior Selection

How should we choose λ?
2 possible solutions:
- Set manually a fixed tightness (λ)
- Thread λ as am **unknow variable** (usually **Gamma distributed**) estimated from the data



$$f(x; k, \theta) = \frac{x^{k-1} e^{-x/\theta}}{\theta^k \Gamma(k)}$$

In this way, BVAR both estimates (e.g. with **Gibbs Sampling**) :
- The Model's coefficients
- The optimal λ that best fits the data

The Project

Project Outline

1. Data cleaning and pre-processing
2. Model Configuration with different parameters, priors and lags.
3. Model building
4. Evaluation and model comparison
5. Comparison with AR and VAR
6. Forecasting with the best configuration
7. Conclusion and discussion

1. Pre-processing

Data Cleaning, Variable Creation, Stationarity

$$inflation = \frac{\mathrm{cpi}_t - \mathrm{cpi}_{t-1}}{\mathrm{cpi}_{t-1}}$$

GDP growth through stationarity Transformation

To evaluate the different models we use data up to 2019 due to Covid

Stationarity

Precondition for VAR models is stationarity.
Log differencing for non rate variables [4, 5]: $\check{y}_{i,t} = \log(y_{i,t}) - \log(y_{i,t-1})$
→ ADF Test

What does stationarity mean?

$\Phi(z) = I_N - \Phi_1 z - \ldots - \Phi_p z^p$, I_N is the identity matrix, $|\Phi(z)| = 0 \Rightarrow |z| > 1.$

The VAR process is stationary if $\phi(z)$ if all root lie outside the unit circle [3]

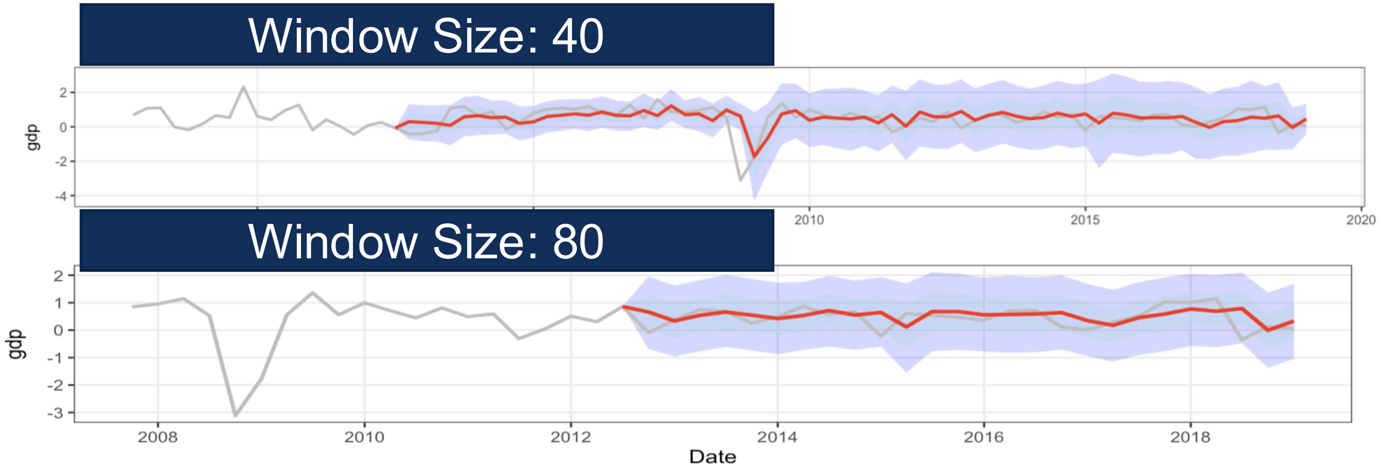
2. Model Configuration
Different parameters, priors and lags

Variable Selection	**Set 0**: GDP, Inflation, Exchange Rate **Set 1:** GDP, Inflation, Exchange Rate, Government Consumption, Employment, Nominal Wage Index, Bent Oil Price, Consumer Prices OECD **Set 2:** GDP, Inflation, Exchange Rate, Government Consumption, Gross Fixed Capital Formation, Exports of goods, Imports of goods, Employement, Unemployment rate accoring to SECO, Nominal Wage Index, Brent Oil Price, Interest Rate, Consumper Prices OECD
Lags Selection	1 quarter, 1 year , 2 years
Prior Selection	Diffusion Prior, Minnesota Prior, Sum of Coefficient Prior

3. Model Building

BVAR Package

Rolling Window Forecast
Window size: 10 years *(we can show the effect, higher variance persistance…)*



Higher Rate of Change in variance and visually larger variance in the mean forecast

4. Evaluatation and Model Comparison

Point Forecasts

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Distribution Forecasts

$$\text{p(x): forecasted prob. dist.}$$

$$y\text{: observed value}$$

$$\text{Log Score} = \log p(y)$$

+ **Benchmark Model** — Comparison to AR(1) forecast

4. Evalutation and Model Comparison

Selection of Trials for GDP Growth → *Similar results for other variables*

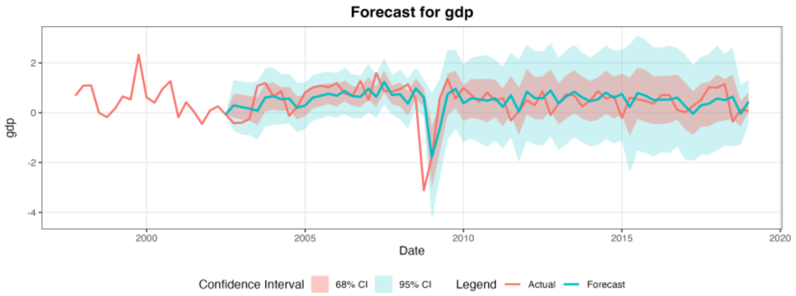
RMSE

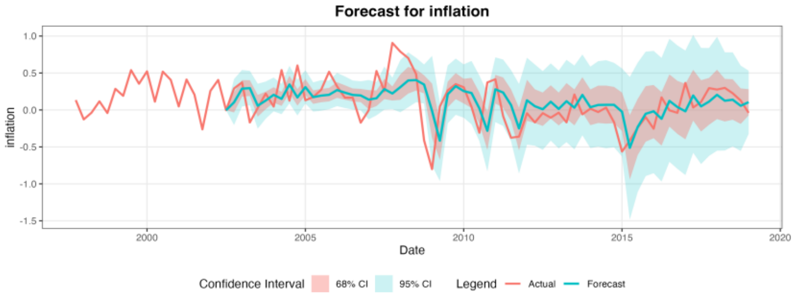
Variation	AR(1) Benchmark model	BVAR Diffusion Prior	BVAR Minnisota Prior	BVAR Sum of Coefficent Prior
Set 0 | Lag 1	0.66	0.678	0.667	0.669
Set 0 | Lag 4		0.885	0.682	0.683
Set 1 | Lag 1		0.779	0.702	0.711
Set 1 | Lag 4		2.80	0.751	0.779
Set 2 | Lag 1		0.93	0.746	0.757
Set 2 | Lag 4		0.47 (window size 80)	0.760	0.856

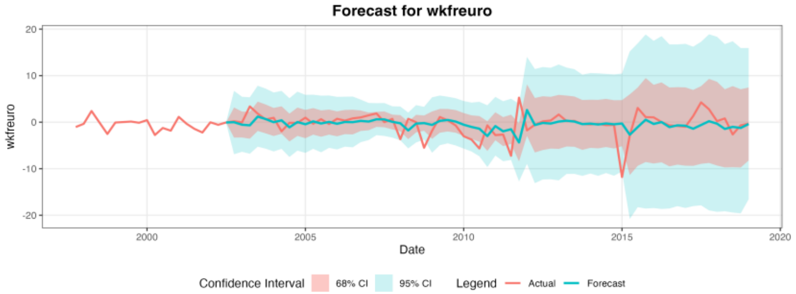
Here you run into issues using OLS → to many variables: impossible estimation with small window size

5. Comparison with AR and VAR

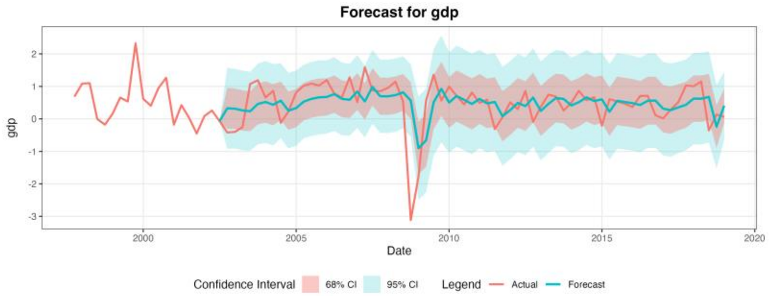
BVAR

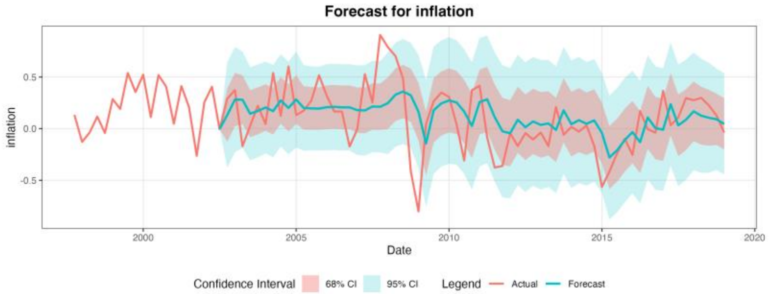


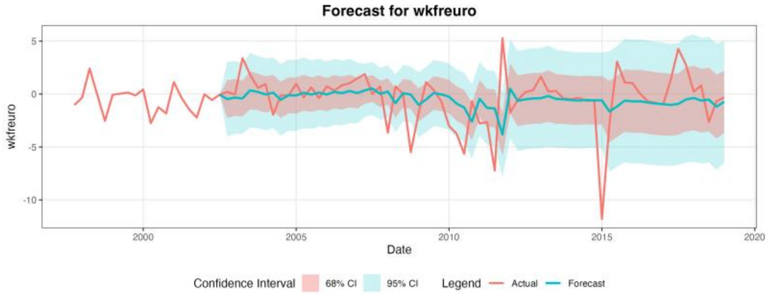




AR(1)





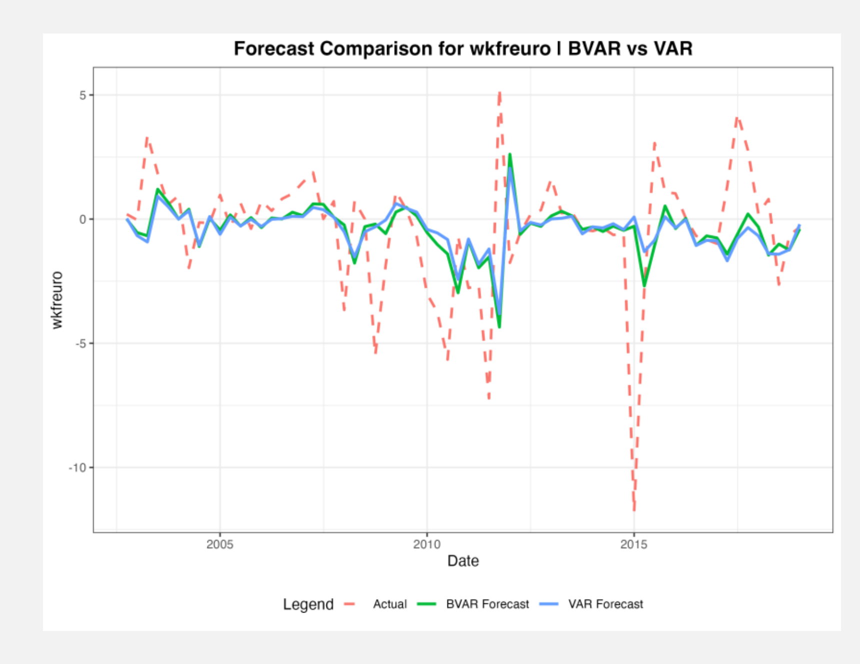


- The AR model has a more consistent variance band around the prediction.
- The confidence interval of the BVAR model is shows larger changes.

5. Comparison with AR and VAR

RMSE

variable	BVAR	VAR	AR(1)
gdp	0.668	0.677	0.659
inflation	0.283	0.290	0.288
wkfreuro	2.658	2.702	2.635



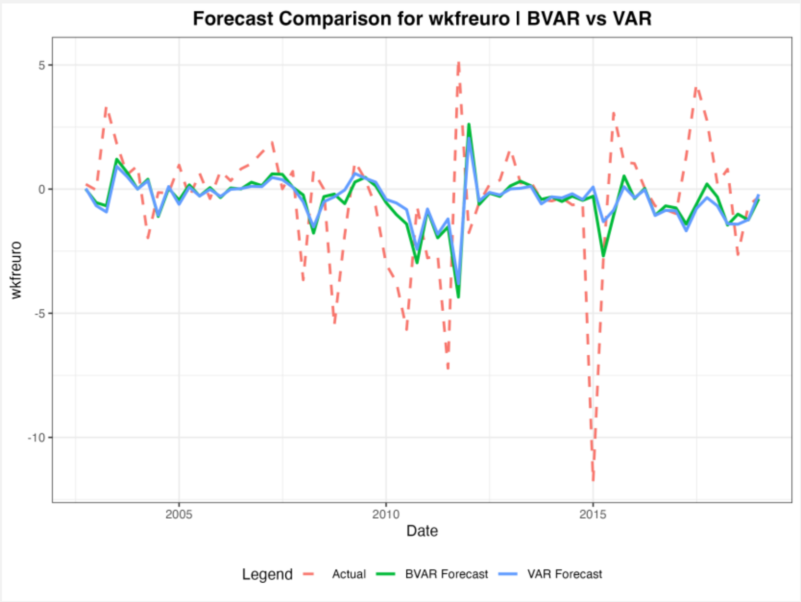
Why is an AR(1) often surprisingly accurate?

- Its simplicity and focus on short-term autocorrelation work well in stable economic environments.
- Short-run relationships between macro variables are often weak or unstable
- More complex models like VAR can introduce extra noise and overfitting without improving forecasts.

5. Comparison with AR and VAR

Mean Log Score

variable	BVAR	AR(1)
gdp	0.500	-1.112
inflation	0.113	-0.217
wkfreuro	-0.468	-2.539



Why do we obtain positive Log Scores?

- The BVAR is very flexible to capture distributions, whereas the AR(1) is penalized for forcing a rigid Gaussian one.
- The BVAR's priors shrink parameters to produce tighter, "sharper" confidence intervals, while the AR(1) produces wide, flat distributions that dilute the density score.

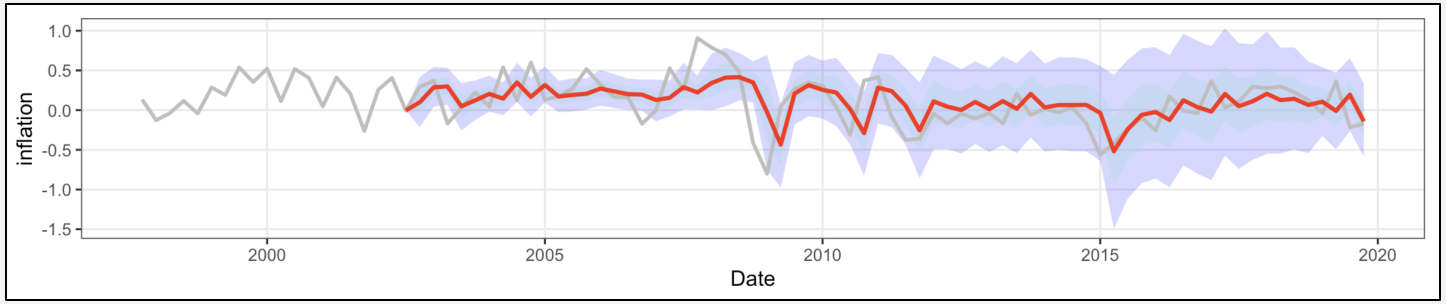
- AR(1) performs better for the point forecast *(lower MRSE)*
- BVAR assigns a higher probability to the actual outcome *(higher Log Score)*

6. Final Model & Performance

Exchange Rate Growth

RMSE: 2.658
Difference to benchmark: -0.023

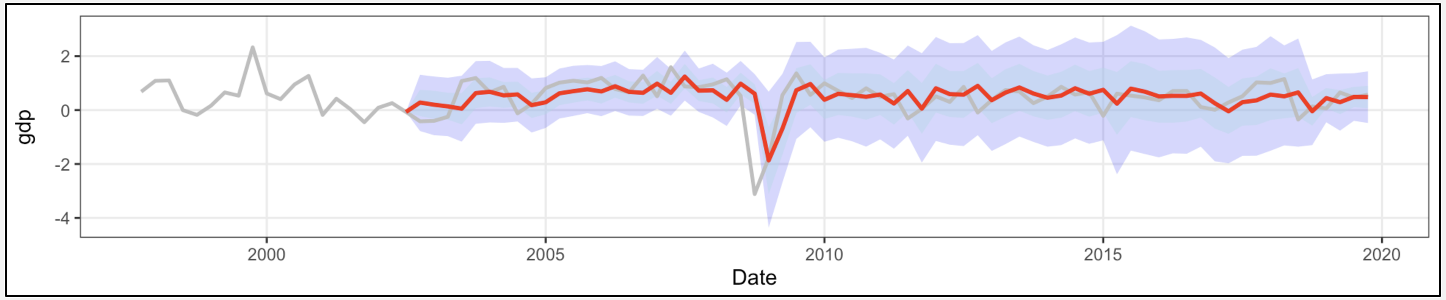
Log Score: -0.4689355



GDP

RMSE: 0.668
Difference to benchmark: -0.01

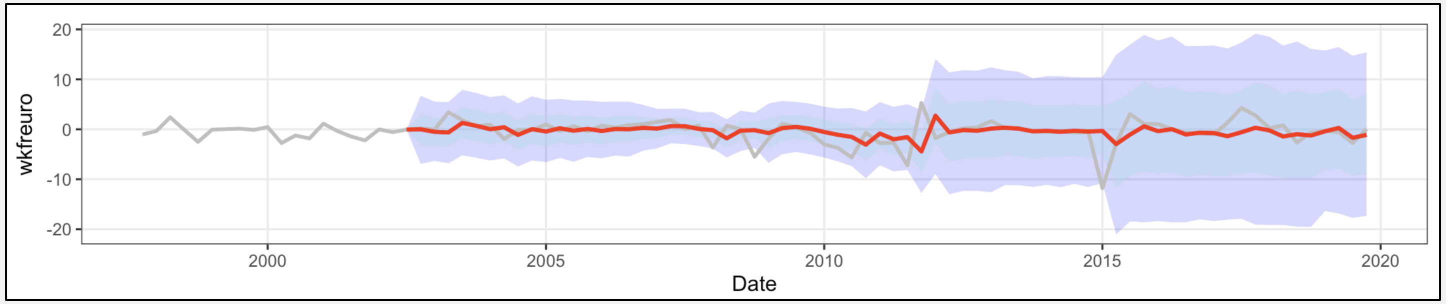
Log Score: 0.5008223



Inflation Rate

RMSE: 0.284
Difference to benchmark: 0.005

Log Score: 0.1139618

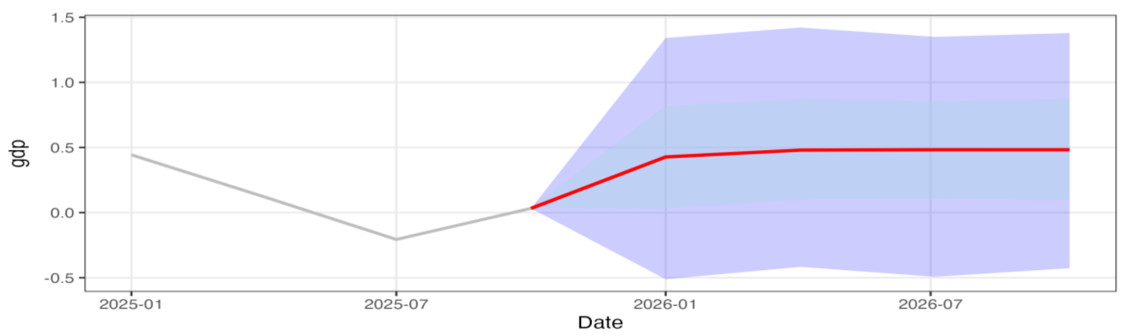


6. Forecasting with the best configuration

GDP Growth

1 Step: **0.427%**

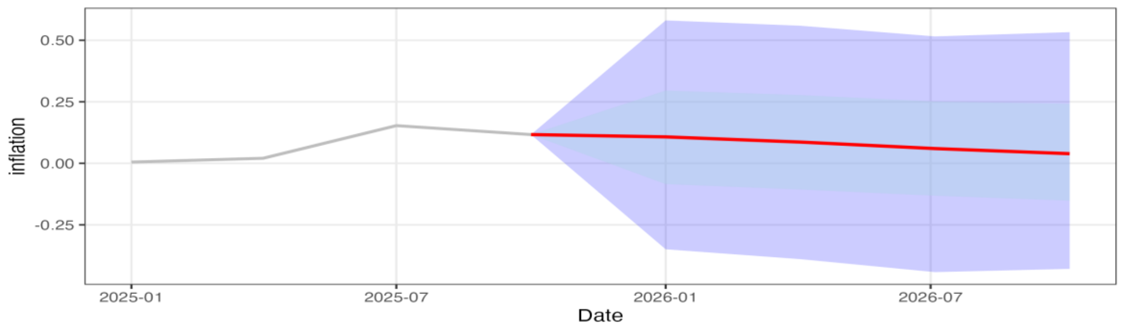
4 Steps: **0.482%**



Inflation Rate

1 Step: **0.107%**

4 Steps: **0.039%**

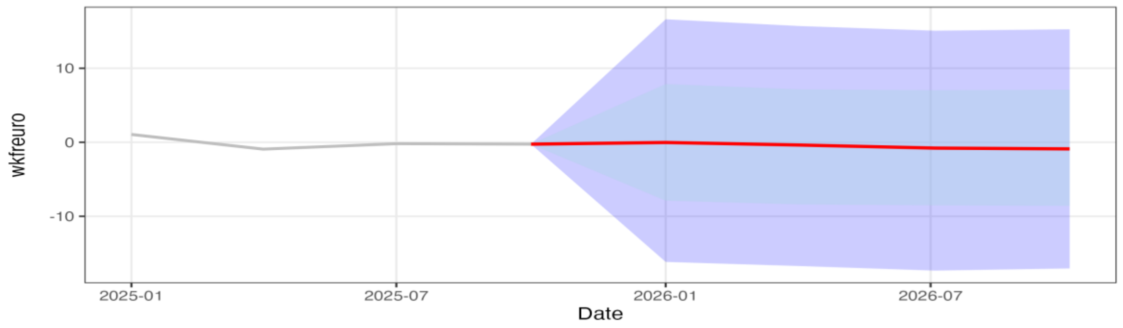


Exchange Rate

From the last observeration transformed using the estimated growth rates

1 Step: **0.914 CHF/€**

4 Steps: **0.895 CHF/€**



6. Forecasting GDP

Comparison with best Configuration and comparison to KOF

GDP Growth	Our Forecast	KOF Forecast
2026-01-01	0.45	0.55
2026-04-01	0.49	0.67
2026-07-01	0.48	0.25
2026-10-01	0.49	0.25

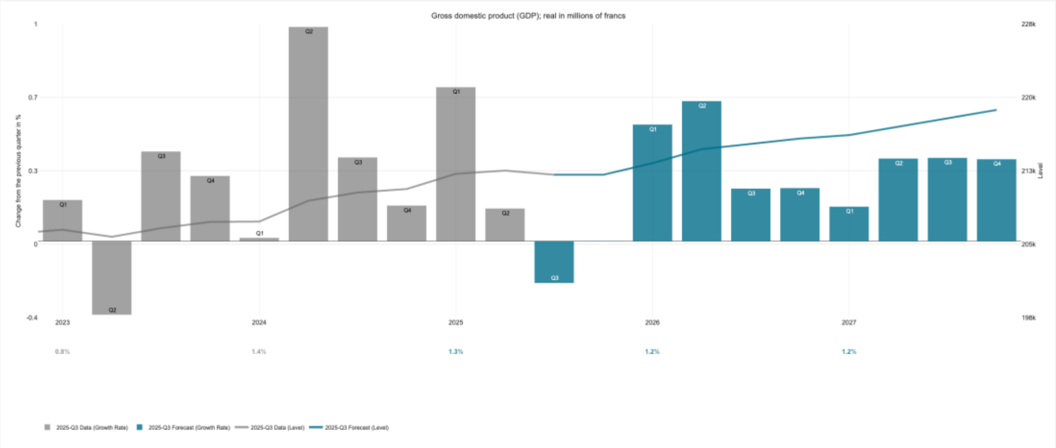


Figure: KOF GDP Forecast [4]

7. Conclusion and discussion
Macroeconomical Interpretation

- **Economic growth (GDP)** is robust and accelerating.
- **Inflation** is rapidly falling (disinflation).
- **Currency Strength** may be the driver, as CHF/EUR confirm its negative trend

Highly favorable scenario for consumers

EUR/CHF 5 years trend

7. Conclusion and discussion
Findings

- BVARs are essential for larger models for managing overparameterization. It consistently outperforms standard OLS VARs, which become unstable with many variables.
- The simple AR(1) model achieved the lowest RMSE for both GDP and the exchange rate, highlighting its power in short-term forecasting.

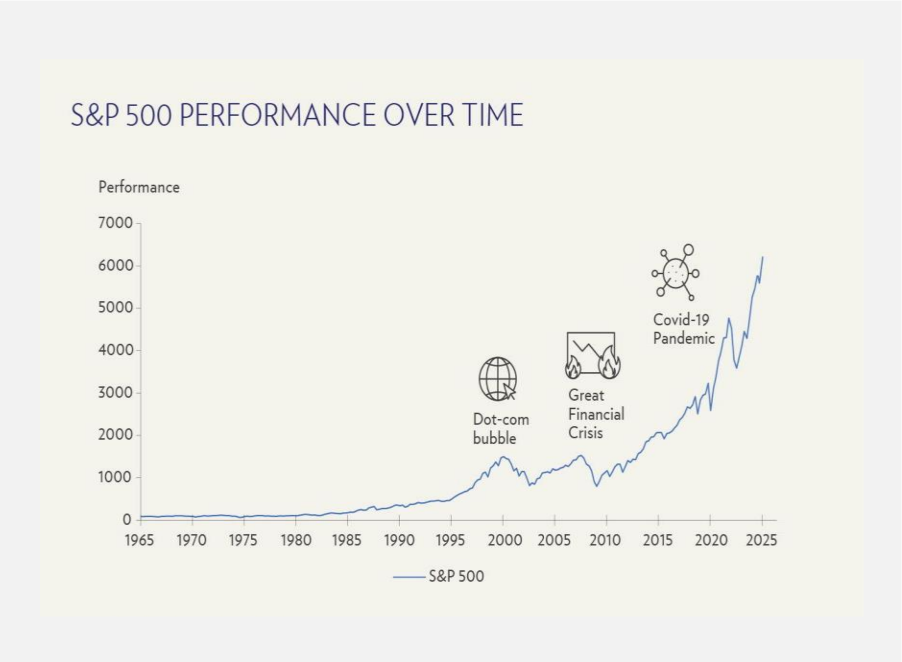
For short horizons, autocorrelation is a stronger signal than complex cross-variable dynamics.

- Within our BVARs, smaller models (fewer variables, 1 lag) with informative priors provided the most robust and stable point forecasts.

7. Conclusion and discussion

Further Considerations and Area of Improvement

- AR(1) performs best in the short term (h = 1, 4), while BVAR models may outperform it over longer horizons as cross-variable effects strengthen.
- Including COVID-period data in the training set could help evaluate its impact on forecast accuracy.
- The project updates data up to 2025 for forecasting but does not account for external shocks or structural breaks (e.g., policy changes, crises). Incorporating external variables and scenario-based forecasts could be a valuable extension.



Sources

1. Good, I. J. “Rational Decisions.” *Journal of the Royal Statistical Society. Series B (Methodological)*, vol. 14, no. 1, 1952, pp. 107–14. *JSTOR*, http://www.jstor.org/stable/2984087. Accessed 18 Nov. 2025.
2. Kuschnig, Nikolas, and Lukas Vashold. "BVAR: Bayesian vector autoregressions with hierarchical prior selection in R." Journal of Statistical Software 100 (2021): 1-27.
3. Hyndman, Rob J., and Athanasopoulos, George, ”Forecasting: Principles and Practice”, 2025, https://otexts.com/fpp2/stationarity.html
4. https://konjunkturprognose.kof.ethz.ch/
5. Bańbura, M., Giannone, D. and Reichlin, L. (2010), Large Bayesian vector auto regressions. J. Appl. Econ., 25: 71-92.